Supplementary Material for "Geometric Algebra-Based Optimal Control with Analytical Derivatives for Robotic Motion Planning"

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Abstract

This document provides detailed information on formulating the optimal control problem with analytical derivatives based on Geometric Algebra (GA). We present the differential action model and cost model constructed using GA, accompanied by their detailed derivation processes. Finally, benchmark results evaluated across various hardware platforms are provided.

NOMENCLATURE

Notation	*Definition*
GA	Geometric algebra.
AD	Automatic differentiation.
DDP	Differential dynamic programming.
SQP	Sequential quadratic programming.
DoF	Degree of freedom.
FD,ID	Forward dynamics and inverse dynamics.
FO,SO	First-, and second-order.

1 Preliminaries

1.1 Geometric Algebra

The core concept of GA is its associative bilinear operator called the geometric product. The geometric product of two vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$ is defined as the sum of the inner product, denoted as $\cdot$ and the outer product, denoted as $\wedge$ of the two vectors

$$\boldsymbol{ab} = \boldsymbol{a} \cdot \boldsymbol{b} + \boldsymbol{a} \wedge \boldsymbol{b}, \quad \forall \boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n \tag{1}$$

The inner product in the equation is a symmetric bilinear form describing the metric property of the space. It can be characterized by a signature (p, q, r) and a basis $\{\boldsymbol{e}_i\}$ of $\mathbb{R}^n$ such that

$$\begin{aligned} &\boldsymbol{e}_i \cdot \boldsymbol{e}_j = 0, \quad (i \neq j) \\ &\boldsymbol{e}_i \cdot \boldsymbol{e}_i = \begin{cases} 1 & 1 \leq i \leq p \\ -1 & p+1 \leq i \leq p+q \\ 0 & p+q+1 \leq i \leq n = p+q+r \end{cases} \end{aligned} \tag{2}$$

The outer product in (1) is the antisymmetric part of the tensor product. The result of the outer product of two vectors is an element in a new vector space, denoted as $\bigwedge^2(\mathbb{R}^n)$. The

new vector space is defined as the space of bivectors, also named as the space with grade 2. Correspondingly, the basis of $\bigwedge^2(\mathbb{R}^n)$ is induced from the basis of $\mathbb{R}^n$, denoted as $\{\boldsymbol{e}_{ij} := \boldsymbol{e}_i \wedge \boldsymbol{e}_j, \text{ for } i \neq j (i, j = 1, 2, \ldots, n)\}$. Successively, the vector space with higher grades can be derived, denoted as $\bigwedge^3(\mathbb{R}^n), \ldots, \bigwedge^n(\mathbb{R}^n)$. It then naturally turns out that the geometric product extends the vector space $\mathbb{R}^n$ to the direct sum of the linear space with different grades

$$\mathbb{G}_{p,q,r} = \bigwedge \mathbb{R}^n := \mathbb{R} \oplus \mathbb{R}^n \oplus \bigwedge^2(\mathbb{R}^n) \oplus \cdots \oplus \bigwedge^n(\mathbb{R}^n) \tag{3}$$

An element in the space is called a multivector. The multivector containing only the grade n part is called a pseudoscalar for the space. It is customary to normalize this to a unit pseudoscalar, denoted as $\mathbf{I}_n$. It turns out that the k-vector space is dual to the $(n-k)$-vector space. The duality is called the Poincaré Duality, and its explicit form can be defined as a linear operator

$$\begin{aligned} \mathcal{J} : \bigwedge^k(\mathbb{R}^n) &\longrightarrow \bigwedge^{n-k}(\mathbb{R}^n) \\ \text{s.t. } \mathcal{J}(\boldsymbol{e}_i^k) = \boldsymbol{e}_i^{n-k} &\Longleftrightarrow \boldsymbol{e}_i^k \wedge \boldsymbol{e}_i^{n-k} = \pm \mathbf{I}_n \end{aligned} \tag{4}$$

The regressive product is the dual of the exterior product (respectively corresponding to the "meet" and "join" in the following text).

$$\boldsymbol{a} \vee \boldsymbol{b} := \mathcal{J}(\mathcal{J}(\boldsymbol{a}) \wedge \mathcal{J}(\boldsymbol{b})), \quad \boldsymbol{a}, \boldsymbol{b} \in \mathbb{G}_{p,q,r} \tag{5}$$

The two most widely used variants of GA in robotics are 3D Conformal Geometric Algebra (CGA) $\mathbb{G}_{4,1,0}$ and 3D Plane-based (or Projective) Geometric Algebra (PGA) $\mathbb{G}_{3,0,1}$. In the subsequent sections of this paper, unless stated otherwise, we will adopt the 3D-PGA formulation, as many derivations for 3D-CGA are analogous. The sixteen basis vectors in $\mathbb{G}_{3,0,1}$ are

$$\begin{aligned} \{1, \boldsymbol{e}_0, \boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_{01}, \boldsymbol{e}_{02}, \boldsymbol{e}_{03}, \\ \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}, \boldsymbol{e}_{123}, \boldsymbol{e}_{023}, \boldsymbol{e}_{031}, \boldsymbol{e}_{012}, \mathbf{I}\} \end{aligned} \tag{6}$$

Planes $\boldsymbol{p}$, lines $\boldsymbol{L}$, and points $\boldsymbol{Q}$ have explicit expressions in PGA [1]. They can be represented as vectors, bivectors and trivectors respectively in the corresponding projective space:

$$\begin{aligned} \text{Plane: } \boldsymbol{p} &= -p_d \boldsymbol{e}_0 + p_1 \boldsymbol{e}_1 + p_2 \boldsymbol{e}_2 + p_3 \boldsymbol{e}_3 \\ \text{Line: } \boldsymbol{L} &= m_1 \boldsymbol{e}_{01} + m_2 \boldsymbol{e}_{02} + m_3 \boldsymbol{e}_{03} + l_1 \boldsymbol{e}_{23} + l_2 \boldsymbol{e}_{31} + l_3 \boldsymbol{e}_{12} \\ \text{Point: } \boldsymbol{Q} &= \boldsymbol{e}_{123} - q_1 \boldsymbol{e}_{023} - q_2 \boldsymbol{e}_{031} - q_3 \boldsymbol{e}_{012} \end{aligned} \tag{7}$$

Any geometric element $\boldsymbol{X}$, such as a plane, line, or point, can have its rigid body motion described uniformly. Such motion can be achieved by twice reflections with two different planes. Suppose two different planes $\boldsymbol{a}$ and $\boldsymbol{b}$ are given, and the simplest rigid body motion of a multivector $\boldsymbol{X}$ in GA is then

$$\boldsymbol{M}_{\boldsymbol{ab}}(\boldsymbol{X}) = \boldsymbol{baXab} \tag{8}$$

The operation $\boldsymbol{M}_{\boldsymbol{ab}}$ is used to describe rigid body motion and is referred to as motors in GA. Define the motor as $\boldsymbol{M} = \boldsymbol{ab}$ with its reverse[1] $\widetilde{\boldsymbol{M}} = \boldsymbol{ba}$. The definition of $\boldsymbol{M}$ naturally satisfies $\widetilde{\boldsymbol{M}}\boldsymbol{M} = \boldsymbol{M}\widetilde{\boldsymbol{M}} = 1$. Therefore, the rigid body motion of a multivector $\boldsymbol{X}$ in GA is represented as:

$$\boldsymbol{X} = \boldsymbol{M}_{\boldsymbol{ab}}(\boldsymbol{X}) = \widetilde{\boldsymbol{M}}\boldsymbol{X}\boldsymbol{M} \tag{9}$$

[1]The reverse operator $\widetilde{}$ in GA denotes reversing the order of geometric products.

1.2 Motor Group

Indeed, motors constitute a group, and it can be verified that they form a Lie group $\mathcal{M}$. Both the group multiplication operation (i.e., the geometric product in GA) and the inversion operation (i.e., the reverse operator in GA) are C^∞-smooth. It is worth noting that the motor group is isomorphic to unit dual quaternions. Both $\mathbb{G}_{4,1,0}$ and $\mathbb{G}_{3,0,1}$ contain the motor group $\mathcal{M}$ as a subalgebra, though with slightly different basis representations. The commonly used representations for these two algebras are provided below.

$$\begin{aligned} \mathcal{M}_{4,1,0} &\subset \operatorname{span}\{1, \boldsymbol{e}_{\infty 1}, \boldsymbol{e}_{\infty 2}, \boldsymbol{e}_{\infty 3}, \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}, \boldsymbol{e}_{\infty 123}\} \\ \mathcal{M}_{3,0,1} &\subset \operatorname{span}\{1, \boldsymbol{e}_{01}, \boldsymbol{e}_{02}, \boldsymbol{e}_{03}, \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}, \boldsymbol{e}_{0123}\} \end{aligned} \tag{10}$$

The Lie algebra $\mathbf{m}$ corresponding to the motor group is the bivector space.

$$\begin{aligned} \mathbf{m}_{4,1,0} &= \operatorname{span}\{\boldsymbol{e}_{\infty 1}, \boldsymbol{e}_{\infty 2}, \boldsymbol{e}_{\infty 3}, \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}\} \\ \mathbf{m}_{3,0,1} &= \operatorname{span}\{\boldsymbol{e}_{01}, \boldsymbol{e}_{02}, \boldsymbol{e}_{03}, \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}\} \end{aligned} \tag{11}$$

The exponential map on the Lie group $\mathcal{M}$ is defined as follows:

$$\exp(\boldsymbol{B}) = e^{\boldsymbol{B}} = \sum_{k=0}^{\infty} \frac{\boldsymbol{B}^k}{k!}, \quad \forall \boldsymbol{B} \in \mathbf{m} \tag{12}$$

For any motor $\boldsymbol{M}$ in the group, it satisfies the property $\boldsymbol{M}\widetilde{\boldsymbol{M}} = 1$. Differentiating this identity yields $\dot{\boldsymbol{M}}\widetilde{\boldsymbol{M}} + \boldsymbol{M}\dot{\widetilde{\boldsymbol{M}}} = 0$. Consider a trajectory of the multivector $\boldsymbol{X}(t) = \widetilde{\boldsymbol{M}}\boldsymbol{X}^0\boldsymbol{M}$ parameterized with t. Taking derivatives on $\boldsymbol{X}(t)$, we have

$$\begin{aligned} \dot{\boldsymbol{X}}(t) &= \dot{\widetilde{\boldsymbol{M}}}\boldsymbol{X}^0\boldsymbol{M} + \widetilde{\boldsymbol{M}}\boldsymbol{X}^0\dot{\boldsymbol{M}} + \widetilde{\boldsymbol{M}}\dot{\boldsymbol{X}}^0\boldsymbol{M} \\ &= \dot{\widetilde{\boldsymbol{M}}}\boldsymbol{M}\boldsymbol{X} + \boldsymbol{X}\widetilde{\boldsymbol{M}}\dot{\boldsymbol{M}} + \widetilde{\boldsymbol{M}}\dot{\boldsymbol{X}}^0\boldsymbol{M} \\ &= -\widetilde{\boldsymbol{M}}\dot{\boldsymbol{M}}\boldsymbol{X} + \boldsymbol{X}\widetilde{\boldsymbol{M}}\dot{\boldsymbol{M}} + \widetilde{\boldsymbol{M}}\dot{\boldsymbol{X}}^0\boldsymbol{M} \\ &= \boldsymbol{X} \times (2\widetilde{\boldsymbol{M}}\dot{\boldsymbol{M}}) + \widetilde{\boldsymbol{M}}\dot{\boldsymbol{X}}^0\boldsymbol{M} \end{aligned} \tag{13}$$

Here, the commutator product $\times$ in the theory of GA is introduced

$$\boldsymbol{X}_1 \times \boldsymbol{X}_2 := \frac{1}{2}(\boldsymbol{X}_1\boldsymbol{X}_2 - \boldsymbol{X}_2\boldsymbol{X}_1), \forall \boldsymbol{X}_1, \boldsymbol{X}_2 \in \mathbb{G}_{p,q,r} \tag{14}$$

In the last line of Eq. (13), the term $2\widetilde{\boldsymbol{M}}\dot{\boldsymbol{M}}$ turns out to be the rigid body velocity with respect to the parameter t expressed in the spatial frame, denoted as $\boldsymbol{V}^s$. It is a bivector in the tangent space of $\mathcal{M}$ at the identity element 1. An important algebraic Lie bracket $[\cdot, \cdot]$ can be defined on the vector space $\mathbf{m}$. It is straightforward to verify, based on the properties of the geometric product, that this operator satisfies the Jacobi identity and other requisite conditions.

$$\begin{aligned} [\boldsymbol{B}_1, \boldsymbol{B}_2] &= \boldsymbol{B}_1\boldsymbol{B}_2 - \boldsymbol{B}_2\boldsymbol{B}_1 \\ &= 2\boldsymbol{B}_1 \times \boldsymbol{B}_2, \quad \forall \boldsymbol{B}_1, \boldsymbol{B}_2 \in \mathbf{m} \end{aligned} \tag{15}$$

For every $\boldsymbol{M} \in \mathcal{M}$, the adjoint action, $\mathrm{Ad}_{\boldsymbol{M}} : \mathbf{m} \to \mathbf{m}$, is a Lie algebra isomorphism that enables change of frames. Superscripts denote reference frames: s for spatial frame (or world frame), b for body frame.

$$\boldsymbol{V}^b = \mathrm{Ad}_{\boldsymbol{M}}(\boldsymbol{V}^s) = \boldsymbol{M}\boldsymbol{V}^s\widetilde{\boldsymbol{M}}, \forall \boldsymbol{M} \in \mathcal{M}, \boldsymbol{V}^s, \boldsymbol{V}^b \in \mathbf{m} \tag{16}$$

1.3 Differantial Kinematics

Here, we directly present the conclusion; interested readers are referred to [2] for further details.

$$\begin{aligned}
\boldsymbol{L}_k &= \tilde{\boldsymbol{M}}_{\lambda(k)} \boldsymbol{L}_k^0 \boldsymbol{M}_{\lambda(k)} \\
\dot{\boldsymbol{L}}_k &= \boldsymbol{L}_k \times \boldsymbol{V}_k \\
\ddot{\boldsymbol{L}}_k &= \dot{\boldsymbol{L}}_k \times \boldsymbol{V}_k + \boldsymbol{L}_k \times \dot{\boldsymbol{V}}_k
\end{aligned} \tag{17}$$

$$\begin{aligned}
\boldsymbol{V}_k &= \sum_{m \preceq k} \dot{q}_m \boldsymbol{L}_m \\
\dot{\boldsymbol{V}}_k &= \sum_{m \preceq k} \ddot{q}_m \boldsymbol{L}_m + \dot{q}_m \dot{\boldsymbol{L}}_m
\end{aligned} \tag{18}$$

1.4 The Dynamics Based on GA

According to Newtonian mechanics, momentum is the product of the mass and velocity of an object. In the PGA, a mass particle is modeled as a Euclidean point $\boldsymbol{Q}$ with a weight m. When the particle is attached to a rigid body, the velocity of $m\boldsymbol{Q}$ is $\dot{\boldsymbol{Q}}$, which equals $\boldsymbol{Q} \times \boldsymbol{V}$. The momentum of the mass particle is then modeled as the joining line of $m\boldsymbol{Q}$ and $\dot{\boldsymbol{Q}}$. It is modeled in the PGA as

$$\boldsymbol{\Pi} = m\boldsymbol{Q} \vee \dot{\boldsymbol{Q}} = m\boldsymbol{Q} \vee (\boldsymbol{Q} \times \boldsymbol{V}) \tag{19}$$

The force acting on the mass particle is modeled as the line determined by the 3D force vector $\boldsymbol{f}$ and the acting point $\boldsymbol{Q}$ [3]. It is named as a wrench and has the following form

$$\boldsymbol{w} = \boldsymbol{Q} \vee \boldsymbol{f} \tag{20}$$

Newton's second law, in modern form, states that the time derivative of the momentum is the force acting on the mass particle:

$$\boldsymbol{w} = \dot{\boldsymbol{\Pi}} = m\boldsymbol{Q} \vee \ddot{\boldsymbol{Q}} \tag{21}$$

For a rigid body, the momentum of the rigid body is defined as the integral of all mass particles contained in it:

$$\boldsymbol{\Pi} = \int_{\Omega} [\rho \boldsymbol{Q} \vee (\boldsymbol{Q} \times \boldsymbol{V})] \, \mathrm{d}\boldsymbol{Q} \tag{22}$$

Ω is the set containing all mass particles in the rigid body and ρ is mass density, a function about $\boldsymbol{Q}$. We recognize that the momentum is linear with respect to the velocity bivector $\boldsymbol{V}$. We define the total inertia map $\boldsymbol{I}(\cdot)$ as a linear function

$$\boldsymbol{I}(\cdot) = \int_{\Omega} [\rho \boldsymbol{Q} \vee (\boldsymbol{Q} \times \cdot)] \, \mathrm{d}\boldsymbol{Q} \tag{23}$$

The inertial duality map takes a bivector velocity $\boldsymbol{V}$ used to describe the kinematics of a motor, and uses the mass distribution of the rigid body to map it to a dual bivector $\boldsymbol{\Pi}$, representing a momentum that dynamical forces can act on. If $\boldsymbol{I}(\cdot)$ is expanded with respect to the basis of the bivector space, an inertia matrix similar to SVA and POE methods is obtained

$$\underline{\boldsymbol{I}} = \begin{bmatrix} m\underline{\hat{\boldsymbol{c}}}^{\mathrm{T}} & m\underline{\boldsymbol{E}}_3 \\ \underline{\boldsymbol{J}} + m\underline{\hat{\boldsymbol{c}}}\underline{\hat{\boldsymbol{c}}}^{\mathrm{T}} & m\underline{\hat{\boldsymbol{c}}} \end{bmatrix} \tag{24}$$

m represents the mass of the rigid body, $\underline{\boldsymbol{E}}_3$ is the 3×3 identity matrix, $\underline{\hat{\boldsymbol{c}}}$ is the skew-symmetric matrix representation generated from the position vector of the reference coordinate frame

relative to the center of mass. $\underline{\boldsymbol{J}}$ is the inertia matrix with respect to the center-of-mass coordinate frame. These definitions are consistent with classical mechanics.

Suppose that the spatial inertia of the rigid body at the initial configuration is $\boldsymbol{I}(\cdot)$ and its coordinates w.r.t. the body-fixed frame $\underline{\boldsymbol{I}}_b$ is known. Then the momentum of the rigid body after a rigid motion $\boldsymbol{M}$ with the velocity $\boldsymbol{V}^s$ is

$$\begin{aligned}\boldsymbol{\Pi}^s &= \int_\Omega \left[\rho(\widetilde{\boldsymbol{M}}\boldsymbol{Q}^0\boldsymbol{M}) \vee ((\widetilde{\boldsymbol{M}}\boldsymbol{Q}^0\boldsymbol{M}) \times \boldsymbol{V}^s)\right] \mathrm{d}(\widetilde{\boldsymbol{M}}\boldsymbol{Q}^0\boldsymbol{M}) \\ &= \widetilde{\boldsymbol{M}}\left[\int_\Omega \rho\boldsymbol{Q}^0 \vee (\boldsymbol{Q}^0 \times (\boldsymbol{M}\boldsymbol{V}^s\widetilde{\boldsymbol{M}}))\mathrm{d}\boldsymbol{Q}^0\right]\boldsymbol{M} \\ &= \widetilde{\boldsymbol{M}}\boldsymbol{I}_b(\boldsymbol{M}\boldsymbol{V}^s\widetilde{\boldsymbol{M}})\boldsymbol{M} = \widetilde{\boldsymbol{M}}\boldsymbol{I}_b(\boldsymbol{V}^b)\boldsymbol{M}\end{aligned} \tag{25}$$

where $\boldsymbol{Q}^0$ is the initial position of $\boldsymbol{Q}$. Denote $\boldsymbol{I}_b(\boldsymbol{V}^b)$ as $\boldsymbol{\Pi}_b$, i.e., the momentum expressed in the body-fixed coordinate frame. Then equation (25) is simplified as

$$\boldsymbol{\Pi}^s = \widetilde{\boldsymbol{M}}\boldsymbol{\Pi}^b\boldsymbol{M} \tag{26}$$

The derivative of the momentum is formulated, then Newtonian dynamics is constructed with reference to Equation (21).

$$\begin{aligned}\boldsymbol{w}^s &= \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Pi}^s = \frac{\mathrm{d}}{\mathrm{d}t}(\widetilde{\boldsymbol{M}}\boldsymbol{I}_b(\boldsymbol{V}^b)\boldsymbol{M}) \\ &= (\widetilde{\boldsymbol{M}}\boldsymbol{I}_b(\boldsymbol{V}^b)\boldsymbol{M}) \times \boldsymbol{V}^s + \widetilde{\boldsymbol{M}}\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{I}_b(\boldsymbol{V}^b)\boldsymbol{M} \\ &= \widetilde{\boldsymbol{M}}\left[\boldsymbol{I}_b(\boldsymbol{V}^b) \times \boldsymbol{V}^b + \boldsymbol{I}_b(\dot{\boldsymbol{V}}^b)\right]\boldsymbol{M}\end{aligned} \tag{27}$$

where $\boldsymbol{w}^s$ is the total wrench in the spatial frame. In the body frame, since $\boldsymbol{w}^b = \boldsymbol{M}\boldsymbol{w}^s\widetilde{\boldsymbol{M}}$, a compact dynamic equation can be constructed.

$$\boldsymbol{w}^b = \dot{\boldsymbol{\Pi}}^b = \boldsymbol{I}_b(\dot{\boldsymbol{V}}^b) + \boldsymbol{I}_b(\boldsymbol{V}^b) \times \boldsymbol{V}^b \tag{28}$$

where, the quantities $\boldsymbol{I}_i^C = \sum_{m \succeq i} \boldsymbol{I}_m, \dot{\boldsymbol{\Pi}}_i^C = \sum_{m \succeq i} \dot{\boldsymbol{\Pi}}_m$ are the quantities defined for an entire sub-tree.

2 First Order Partial Derivatives of ID

In this appendix we provide a complete derivation of the first- and second-order derivatives of

$$\dot{\boldsymbol{\Pi}}_k = \boldsymbol{I}_k(\dot{\boldsymbol{V}}_k) + \boldsymbol{\Pi}_k \times \boldsymbol{V}_k \tag{29}$$

$$\tau_k = \boldsymbol{L}_k * \left(\sum_{m \succeq k} \dot{\boldsymbol{\Pi}}_m\right) = \boldsymbol{L}_k * \dot{\boldsymbol{\Pi}}_k^C \tag{30}$$

2.1 Assumptions and basic identities

We start from the identities [4]

Derivatives of $\boldsymbol{I}_k$.

$$\frac{\partial \boldsymbol{I}_k(\cdot)}{\partial q_i} = \begin{cases} \boldsymbol{I}_k(\boldsymbol{L}_i \times \cdot) + \boldsymbol{I}_k(\cdot) \times \boldsymbol{L}_i & \text{if } i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{31}$$

Derivatives of $\boldsymbol{L}_k$.

$$\frac{\partial \boldsymbol{L}_k}{\partial q_i} = \begin{cases} \boldsymbol{L}_k \times \boldsymbol{L}_i & \text{if } i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{32}$$

Derivatives of $\boldsymbol{V}_k$.

$$\frac{\partial \boldsymbol{V}_k}{\partial q_i} = \begin{cases} \boldsymbol{V}_k \times \boldsymbol{L}_i + \dot{\boldsymbol{L}}_i & \text{if } i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{33}$$

$$\frac{\partial \boldsymbol{V}_k}{\partial \dot{q}_i} = \begin{cases} \boldsymbol{L}_i & \text{if } i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{34}$$

Derivatives of $\dot{\boldsymbol{L}}_k$.

$$\frac{\partial \dot{\boldsymbol{L}}_k}{\partial q_i} = \begin{cases} \dot{\boldsymbol{L}}_k \times \boldsymbol{L}_i + \boldsymbol{L}_k \times \dot{\boldsymbol{L}}_i & \text{if } i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{35}$$

$$\frac{\partial \dot{\boldsymbol{L}}_k}{\partial \dot{q}_i} = \begin{cases} \boldsymbol{L}_k \times \boldsymbol{L}_i & \text{if } i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{36}$$

Derivatives of $\dot{\boldsymbol{V}}_k$.

$$\frac{\partial \dot{\boldsymbol{V}}_k}{\partial q_i} = \begin{cases} \dot{\boldsymbol{V}}_k \times \boldsymbol{L}_i + \boldsymbol{V}_k \times \dot{\boldsymbol{L}}_i + \ddot{\boldsymbol{L}}_i & \text{if } i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{37}$$

$$\frac{\partial \dot{\boldsymbol{V}}_k}{\partial \dot{q}_i} = \begin{cases} \boldsymbol{V}_k \times \boldsymbol{L}_i + 2\dot{\boldsymbol{L}}_i & \text{if } i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{38}$$

$$\frac{\partial \dot{\boldsymbol{V}}_k}{\partial \ddot{q}_i} = \begin{cases} \boldsymbol{L}_i & \text{if } i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{39}$$

Derivatives of $\ddot{\boldsymbol{L}}_k$.

$$\frac{\partial \ddot{\boldsymbol{L}}_k}{\partial q_i} = \begin{cases} \ddot{\boldsymbol{L}}_k \times \boldsymbol{L}_i + 2\dot{\boldsymbol{L}}_k \times \dot{\boldsymbol{L}}_i + \boldsymbol{L}_k \times \ddot{\boldsymbol{L}}_i & \text{if } i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{40}$$

$$\frac{\partial \ddot{\boldsymbol{L}}_k}{\partial \dot{q}_i} = \begin{cases} 2(\dot{\boldsymbol{L}}_k \times \boldsymbol{L}_i + \boldsymbol{L}_k \times \dot{\boldsymbol{L}}_i) & \text{if } i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{41}$$

$$\frac{\partial \ddot{\boldsymbol{L}}_k}{\partial \ddot{q}_i} = \begin{cases} \boldsymbol{L}_k \times \boldsymbol{L}_i & \text{if } i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{42}$$

We also use the Jacobi identity

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) + \boldsymbol{c} \times (\boldsymbol{a} \times \boldsymbol{b}) + \boldsymbol{b} \times (\boldsymbol{c} \times \boldsymbol{a}) = 0.$$

2.2 Definition of the covariant derivative

For convenience, define the q_i-direction covariant derivative

$$\mathcal{D}_i \boldsymbol{X} := \boldsymbol{L}_i \times \boldsymbol{X} + \frac{\partial \boldsymbol{X}}{\partial q_i}. \tag{43}$$

The choice (43) is natural because, using (32),

$$\mathcal{D}_i \boldsymbol{L}_k = \boldsymbol{L}_i \times \boldsymbol{L}_k + \frac{\partial \boldsymbol{L}_k}{\partial q_i} = [\boldsymbol{L}_i, \boldsymbol{L}_k] + [\boldsymbol{L}_k, \boldsymbol{L}_i] = 0.$$

Hence,

$$\mathcal{D}_i \boldsymbol{L}_k = 0. \tag{44}$$

Moreover, using (31),

$$\begin{aligned}
\mathcal{D}_i(\boldsymbol{I}_k(\boldsymbol{X})) &= \boldsymbol{L}_i \times \boldsymbol{I}_k(\boldsymbol{X}) + \frac{\partial}{\partial q_i} \boldsymbol{I}_k(\boldsymbol{X}) \\
&= \boldsymbol{L}_i \times \boldsymbol{I}_k(\boldsymbol{X}) + \left(\frac{\partial \boldsymbol{I}_k}{\partial q_i}\right)(\boldsymbol{X}) + \boldsymbol{I}_k\left(\frac{\partial \boldsymbol{X}}{\partial q_i}\right) \\
&= \boldsymbol{L}_i \times \boldsymbol{I}_k(\boldsymbol{X}) + \boldsymbol{I}_k(\boldsymbol{L}_i \times \boldsymbol{X}) - \boldsymbol{L}_i \times \boldsymbol{I}_k(\boldsymbol{X}) + \boldsymbol{I}_k\left(\frac{\partial \boldsymbol{X}}{\partial q_i}\right) \\
&= \boldsymbol{I}_k\left(\boldsymbol{L}_i \times \boldsymbol{X} + \frac{\partial \boldsymbol{X}}{\partial q_i}\right) \\
&= \boldsymbol{I}_k(\mathcal{D}_i \boldsymbol{X}).
\end{aligned} \tag{45}$$

Therefore,

$$\boxed{\mathcal{D}_i(\boldsymbol{I}_k(\boldsymbol{X})) = \boldsymbol{I}_k(\mathcal{D}_i \boldsymbol{X}), \quad (i \preceq k)} \tag{46}$$

2.3 Preliminary identities

Lemma 1 (Covariant derivatives of the basic quantities)**.** *The following identities hold:*

$$\mathcal{D}_i \boldsymbol{L}_k = \begin{cases} 0 & i \preceq k \\ \boldsymbol{L}_i \times \boldsymbol{L}_k & \textit{otherwise} \end{cases} \tag{47}$$

$$\mathcal{D}_i \boldsymbol{V}_k = \begin{cases} \dot{\boldsymbol{L}}_i & i \preceq k \\ \boldsymbol{L}_i \times \boldsymbol{V}_k & \textit{otherwise} \end{cases} \tag{48}$$

$$\mathcal{D}_i \dot{\boldsymbol{L}}_k = \begin{cases} \boldsymbol{L}_k \times \dot{\boldsymbol{L}}_i & i \preceq k \\ \boldsymbol{L}_i \times \dot{\boldsymbol{L}}_k & \textit{otherwise} \end{cases} \tag{49}$$

$$\mathcal{D}_i \dot{\boldsymbol{V}}_k = \begin{cases} \boldsymbol{V}_k \times \dot{\boldsymbol{L}}_i + \ddot{\boldsymbol{L}}_i & i \preceq k \\ \boldsymbol{L}_i \times \dot{\boldsymbol{V}}_k & \textit{otherwise} \end{cases} \tag{50}$$

$$\mathcal{D}_i \ddot{\boldsymbol{L}}_k = \begin{cases} 2\dot{\boldsymbol{L}}_k \times \dot{\boldsymbol{L}}_i + \boldsymbol{L}_k \times \ddot{\boldsymbol{L}}_i & i \preceq k \\ \boldsymbol{L}_i \times \ddot{\boldsymbol{L}}_k & \textit{otherwise} \end{cases} \tag{51}$$

2.4 A useful bilinear operator

Define

$$\mathcal{B}_{k,\boldsymbol{v}}(\boldsymbol{X}) := \boldsymbol{I}_k(\boldsymbol{v} \times \boldsymbol{X}) + \boldsymbol{I}_k(\boldsymbol{X}) \times \boldsymbol{v} + \boldsymbol{I}_k(\boldsymbol{v}) \times \boldsymbol{X}. \tag{52}$$

Composite term for $\mathcal{B}^C_{k,\boldsymbol{v}}(\boldsymbol{X})$ is

$$\mathcal{B}^C_{k,\boldsymbol{v}}(\boldsymbol{X}) = \boldsymbol{I}^C_k(\boldsymbol{v} \times \boldsymbol{X}) + \boldsymbol{I}^C_k(\boldsymbol{X}) \times \boldsymbol{v} + \boldsymbol{I}^C_k(\boldsymbol{v}) \times \boldsymbol{X}. \tag{53}$$

Lemma 2 (Algebraic symmetry identity)**.** *For any Lie-algebra-valued quantities* $\boldsymbol{a}, \boldsymbol{b}$*,*

$$\mathcal{B}_{k,\boldsymbol{a}}(\boldsymbol{b}) - \mathcal{B}_{k,\boldsymbol{b}}(\boldsymbol{a}) = 2\boldsymbol{I}_k(\boldsymbol{a} \times \boldsymbol{b}). \tag{54}$$

Equivalently,

$$\mathcal{B}_{k,\boldsymbol{b}}(\boldsymbol{a}) + 2\boldsymbol{I}_k(\boldsymbol{a} \times \boldsymbol{b}) = \mathcal{B}_{k,\boldsymbol{a}}(\boldsymbol{b}). \tag{55}$$

Proof. By definition,

$$\mathcal{B}_{k,\boldsymbol{a}}(\boldsymbol{b}) = \boldsymbol{I}_k(\boldsymbol{a} \times \boldsymbol{b}) + \boldsymbol{I}_k(\boldsymbol{b}) \times \boldsymbol{a} + \boldsymbol{I}_k(\boldsymbol{a}) \times \boldsymbol{b},$$

and

$$\mathcal{B}_{k,\boldsymbol{b}}(\boldsymbol{a}) = \boldsymbol{I}_k(\boldsymbol{b} \times \boldsymbol{a}) + \boldsymbol{I}_k(\boldsymbol{a}) \times \boldsymbol{b} + \boldsymbol{I}_k(\boldsymbol{b}) \times \boldsymbol{a}.$$

Subtracting the two expressions gives

$$\mathcal{B}_{k,\boldsymbol{a}}(\boldsymbol{b}) - \mathcal{B}_{k,\boldsymbol{b}}(\boldsymbol{a}) = \boldsymbol{I}_k(\boldsymbol{a} \times \boldsymbol{b}) - \boldsymbol{I}_k(\boldsymbol{b} \times \boldsymbol{a}).$$

Since $\boldsymbol{b} \times \boldsymbol{a} = -(\boldsymbol{a} \times \boldsymbol{b})$, we obtain

$$\mathcal{B}_{k,\boldsymbol{a}}(\boldsymbol{b}) - \mathcal{B}_{k,\boldsymbol{b}}(\boldsymbol{a}) = 2\boldsymbol{I}_k(\boldsymbol{a} \times \boldsymbol{b}).$$

□

Lemma 3 (Derivative rules for $\mathcal{B}_{k,\boldsymbol{v}}(\boldsymbol{X})$)**.** *The following identities hold when* $j \preceq k$*:*

$$\mathcal{D}_j\big(\mathcal{B}_{k,\boldsymbol{v}}(\boldsymbol{X})\big) = \mathcal{B}_{k,\boldsymbol{v}}(\mathcal{D}_j\boldsymbol{X}) + \mathcal{B}_{k,\mathcal{D}_j\boldsymbol{v}}(\boldsymbol{X}), \quad j \preceq k \tag{56}$$

$$\frac{\partial}{\partial \dot{q}_j}\big(\mathcal{B}_{k,\boldsymbol{v}}(\boldsymbol{X})\big) = \mathcal{B}_{k,\boldsymbol{v}}\left(\frac{\partial \boldsymbol{X}}{\partial \dot{q}_j}\right) + \mathcal{B}_{k,\frac{\partial \boldsymbol{v}}{\partial \dot{q}_j}}(\boldsymbol{X}), \quad j \preceq k \tag{57}$$

$$\frac{\partial}{\partial \ddot{q}_j}\big(\mathcal{B}_{k,\boldsymbol{v}}(\boldsymbol{X})\big) = \mathcal{B}_{k,\boldsymbol{v}}\left(\frac{\partial \boldsymbol{X}}{\partial \ddot{q}_j}\right) + \mathcal{B}_{k,\frac{\partial \boldsymbol{v}}{\partial \ddot{q}_j}}(\boldsymbol{X}), \quad j \preceq k \tag{58}$$

Proof. We prove (56); the other two are analogous. Using (52) and (46),

$$\begin{aligned}
\mathcal{D}_j\big(\mathcal{B}_{k,\boldsymbol{v}}(\boldsymbol{X})\big) &= \mathcal{D}_j\big(\boldsymbol{I}_k(\boldsymbol{v} \times \boldsymbol{X})\big) + \mathcal{D}_j\big(\boldsymbol{I}_k(\boldsymbol{X}) \times \boldsymbol{v}\big) + \mathcal{D}_j\big(\boldsymbol{I}_k(\boldsymbol{v}) \times \boldsymbol{X}\big) \\
&= \boldsymbol{I}_k\big(\mathcal{D}_j(\boldsymbol{v} \times \boldsymbol{X})\big) + \mathcal{D}_j(\boldsymbol{I}_k(\boldsymbol{X})) \times \boldsymbol{v} + \boldsymbol{I}_k(\boldsymbol{X}) \times \mathcal{D}_j\boldsymbol{v} + \mathcal{D}_j(\boldsymbol{I}_k(\boldsymbol{v})) \times \boldsymbol{X} + \boldsymbol{I}_k(\boldsymbol{v}) \times \mathcal{D}_j\boldsymbol{X} \\
&= \boldsymbol{I}_k\big((\mathcal{D}_j\boldsymbol{v}) \times \boldsymbol{X} + \boldsymbol{v} \times (\mathcal{D}_j\boldsymbol{X})\big) + \boldsymbol{I}_k(\mathcal{D}_j\boldsymbol{X}) \times \boldsymbol{v} + \boldsymbol{I}_k(\boldsymbol{X}) \times \mathcal{D}_j\boldsymbol{v} \\
&\qquad + \boldsymbol{I}_k(\mathcal{D}_j\boldsymbol{v}) \times \boldsymbol{X} + \boldsymbol{I}_k(\boldsymbol{v}) \times \mathcal{D}_j\boldsymbol{X} \\
&= \mathcal{B}_{k,\boldsymbol{v}}(\mathcal{D}_j\boldsymbol{X}) + \mathcal{B}_{k,\mathcal{D}_j\boldsymbol{v}}(\boldsymbol{X}).
\end{aligned} \tag{59}$$

□

2.5 First-order derivatives of $\dot{\boldsymbol{\Pi}}_k$

Proposition 1 (First-order derivatives of $\dot{\boldsymbol{\Pi}}_k$)**.** *The following identities hold when* $j \preceq k$*:*

$$\mathcal{D}_i\dot{\boldsymbol{\Pi}}_k = \boldsymbol{I}_k(\ddot{\boldsymbol{L}}_i) + \mathcal{B}_{k,\boldsymbol{V}_k}(\dot{\boldsymbol{L}}_i), \quad i \preceq k \tag{60}$$

$$\frac{\partial \dot{\boldsymbol{\Pi}}_k}{\partial \dot{q}_i} = 2\boldsymbol{I}_k(\dot{\boldsymbol{L}}_i) + \mathcal{B}_{k,\boldsymbol{V}_k}(\boldsymbol{L}_i), \quad i \preceq k \tag{61}$$

$$\frac{\partial \dot{\boldsymbol{\Pi}}_k}{\partial \ddot{q}_i} = \boldsymbol{I}_k(\boldsymbol{L}_i), \quad i \preceq k \tag{62}$$

Proof. Starting from (28),

$$\dot{\boldsymbol{\Pi}}_k = \boldsymbol{I}_k(\dot{\boldsymbol{V}}_k) + \boldsymbol{I}_k(\boldsymbol{V}_k) \times \boldsymbol{V}_k.$$

For (60), apply $\mathcal{D}_i$:

$$\begin{aligned}
\mathcal{D}_i\dot{\boldsymbol{\Pi}}_k &= \mathcal{D}_i(\boldsymbol{I}_k(\dot{\boldsymbol{V}}_k)) + \mathcal{D}_i\big([\boldsymbol{I}_k(\boldsymbol{V}_k), \boldsymbol{V}_k]\big) \\
&= \boldsymbol{I}_k(\mathcal{D}_i\dot{\boldsymbol{V}}_k) + [\mathcal{D}_i(\boldsymbol{I}_k(\boldsymbol{V}_k)), \boldsymbol{V}_k] + [\boldsymbol{I}_k(\boldsymbol{V}_k), \mathcal{D}_i\boldsymbol{V}_k] \\
&= \boldsymbol{I}_k(\ddot{\boldsymbol{L}}_i + \boldsymbol{V}_k \times \dot{\boldsymbol{L}}_i) + [\boldsymbol{I}_k(\dot{\boldsymbol{L}}_i), \boldsymbol{V}_k] + [\boldsymbol{I}_k(\boldsymbol{V}_k), \dot{\boldsymbol{L}}_i] \\
&= \boldsymbol{I}_k(\ddot{\boldsymbol{L}}_i) + \boldsymbol{I}_k(\boldsymbol{V}_k \times \dot{\boldsymbol{L}}_i) + \boldsymbol{I}_k(\dot{\boldsymbol{L}}_i) \times \boldsymbol{V}_k + \boldsymbol{I}_k(\boldsymbol{V}_k) \times \dot{\boldsymbol{L}}_i \\
&= \boldsymbol{I}_k(\ddot{\boldsymbol{L}}_i) + \mathcal{B}_{k,\boldsymbol{V}_k}(\dot{\boldsymbol{L}}_i). \qquad (63)
\end{aligned}$$

For (61), differentiate with respect to $\dot{q}_i$:

$$\begin{aligned}
\frac{\partial\dot{\boldsymbol{\Pi}}_k}{\partial\dot{q}_i} &= \boldsymbol{I}_k\left(\frac{\partial\dot{\boldsymbol{V}}_k}{\partial\dot{q}_i}\right) + \left[\boldsymbol{I}_k\left(\frac{\partial\boldsymbol{V}_k}{\partial\dot{q}_i}\right), \boldsymbol{V}_k\right] + \left[\boldsymbol{I}_k(\boldsymbol{V}_k), \frac{\partial\boldsymbol{V}_k}{\partial\dot{q}_i}\right] \\
&= \boldsymbol{I}_k(\boldsymbol{V}_k \times \boldsymbol{L}_i + 2\dot{\boldsymbol{L}}_i) + [\boldsymbol{I}_k(\boldsymbol{L}_i), \boldsymbol{V}_k] + [\boldsymbol{I}_k(\boldsymbol{V}_k), \boldsymbol{L}_i] \\
&= \mathcal{B}_{k,\boldsymbol{V}_k}(\boldsymbol{L}_i) + 2\boldsymbol{I}_k(\dot{\boldsymbol{L}}_i). \qquad (64)
\end{aligned}$$

For (62), differentiate with respect to $\ddot{q}_i$:

$$\begin{aligned}
\frac{\partial\dot{\boldsymbol{\Pi}}_k}{\partial\ddot{q}_i} &= \boldsymbol{I}_k\left(\frac{\partial\dot{\boldsymbol{V}}_k}{\partial\ddot{q}_i}\right) + \left[\boldsymbol{I}_k\left(\frac{\partial\boldsymbol{V}_k}{\partial\ddot{q}_i}\right), \boldsymbol{V}_k\right] + \left[\boldsymbol{I}_k(\boldsymbol{V}_k), \frac{\partial\boldsymbol{V}_k}{\partial\ddot{q}_i}\right] \\
&= \boldsymbol{I}_k(\boldsymbol{L}_i) + 0 + 0 \\
&= \boldsymbol{I}_k(\boldsymbol{L}_i). \qquad (65)
\end{aligned}$$

□

2.6 First-order derivatives of τ_k

The FO Derivative of τ With Respect to q when $i \preceq k$

$$\begin{aligned}
\frac{\partial\tau_k}{\partial q_i} &= \frac{\partial}{\partial q_i}\left(\boldsymbol{L}_k * \sum_{m\succeq k}\dot{\boldsymbol{\Pi}}_m\right) \\
&= \boldsymbol{L}_k * \sum_{m\succeq k}\left(\boldsymbol{L}_i \times \dot{\boldsymbol{\Pi}}_m + \frac{\partial\dot{\boldsymbol{\Pi}}_m}{\partial q_i}\right) \\
&= \boldsymbol{L}_k * \sum_{m\succeq k}(\mathcal{D}_i\dot{\boldsymbol{\Pi}}_m) \\
&= \boldsymbol{L}_k * \sum_{m\succeq k}\Big(\boldsymbol{I}_m(\ddot{\boldsymbol{L}}_i) + \mathcal{B}_{m,\boldsymbol{V}_m}(\dot{\boldsymbol{L}}_i)\Big) \\
&= \boldsymbol{L}_k * \Big(\boldsymbol{I}_k^C(\ddot{\boldsymbol{L}}_i) + \mathcal{B}_{k,\boldsymbol{V}_k}^C(\dot{\boldsymbol{L}}_i)\Big) \qquad (66)
\end{aligned}$$

$$
\begin{aligned}
\frac{\partial \tau_i}{\partial q_k} &= \frac{\partial}{\partial q_k}\left(\boldsymbol{L}_i * \sum_{m \succeq i} \dot{\boldsymbol{\Pi}}_m\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial q_k} \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} (\mathcal{D}_k \dot{\boldsymbol{\Pi}}_m + \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} \Big(\boldsymbol{I}_m(\ddot{\boldsymbol{L}}_k) + \mathcal{B}_{m,\boldsymbol{V}_m}(\dot{\boldsymbol{L}}_k) + \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k\Big) \\
&= \boldsymbol{L}_i * \Big(\boldsymbol{I}_k^C(\ddot{\boldsymbol{L}}_k) + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\dot{\boldsymbol{L}}_k) + \dot{\boldsymbol{\Pi}}_k^C \times \boldsymbol{L}_k\Big)
\end{aligned} \tag{67}
$$

The FO Derivative of τ With Respect to $\dot{q}$ when $i \preceq k$

$$
\begin{aligned}
\frac{\partial \tau_k}{\partial \dot{q}_i} &= \frac{\partial}{\partial \dot{q}_i}\left(\boldsymbol{L}_k * \sum_{m \succeq k} \dot{\boldsymbol{\Pi}}_m\right) \\
&= \boldsymbol{L}_k * \sum_{m \succeq k} \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \dot{q}_i} \\
&= \boldsymbol{L}_k * \sum_{m \succeq k} \Big(\mathcal{B}_{m,\boldsymbol{V}_m}(\boldsymbol{L}_i) + 2\boldsymbol{I}_m(\dot{\boldsymbol{L}}_i)\Big) \\
&= \boldsymbol{L}_k * \Big(\mathcal{B}^C_{k,\boldsymbol{V}_k}(\boldsymbol{L}_i) + 2\boldsymbol{I}_k^C(\dot{\boldsymbol{L}}_i)\Big)
\end{aligned} \tag{68}
$$

$$
\begin{aligned}
\frac{\partial \tau_i}{\partial \dot{q}_k} &= \frac{\partial}{\partial \dot{q}_k}\left(\boldsymbol{L}_i * \sum_{m \succeq i} \dot{\boldsymbol{\Pi}}_m\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \dot{q}_k} \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} \Big(\mathcal{B}_{m,\boldsymbol{V}_m}(\boldsymbol{L}_k) + 2\boldsymbol{I}_m(\dot{\boldsymbol{L}}_k)\Big) \\
&= \boldsymbol{L}_i * \Big(\mathcal{B}^C_{k,\boldsymbol{V}_k}(\boldsymbol{L}_k) + 2\boldsymbol{I}_k^C(\dot{\boldsymbol{L}}_k)\Big)
\end{aligned} \tag{69}
$$

The FO Derivative of τ With Respect to $\ddot{q}$ when $i \preceq k$

$$
\frac{\partial \tau_k}{\partial \ddot{q}_i} = \boldsymbol{L}_k * \sum_{m \succeq k} \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_i} = \boldsymbol{L}_k * \sum_{m \succeq k} \boldsymbol{I}_m(\boldsymbol{L}_i) = \boldsymbol{L}_k * \boldsymbol{I}_k^C(\boldsymbol{L}_i) \tag{70}
$$

$$
\frac{\partial \tau_i}{\partial \ddot{q}_k} = \boldsymbol{L}_i * \sum_{m \succeq i} \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_k} = \boldsymbol{L}_i * \sum_{m \succeq k} \boldsymbol{I}_m(\boldsymbol{L}_k) = \boldsymbol{L}_i * \boldsymbol{I}_k^C(\boldsymbol{L}_k) \tag{71}
$$

3 Second Order Partial Derivatives of ID

3.1 Second-order derivatives of $\dot{\mathbf{\Pi}}_k$

Proposition 2 (Second-order derivatives of $\dot{\mathbf{\Pi}}_k$)**.** *The following identities hold:*

$$\mathcal{D}_i\mathcal{D}_j\dot{\mathbf{\Pi}}_k = \boldsymbol{I}_k(\boldsymbol{L}_j \times \ddot{\boldsymbol{L}}_i) + \mathcal{B}_{k,\boldsymbol{V}_k}({}_j\!\times\dot{\boldsymbol{L}}_i) + \mathcal{B}_{k,\dot{\boldsymbol{L}}_j}(\dot{\boldsymbol{L}}_i), \tag{72}$$

$$\mathcal{D}_i\left(\frac{\partial\dot{\mathbf{\Pi}}_k}{\partial\dot{q}_j}\right) = \mathcal{B}_{k,{}_j}(\dot{\boldsymbol{L}}_i), \tag{73}$$

$$\frac{\partial}{\partial\dot{q}_i}\Big(\mathcal{D}_j\dot{\mathbf{\Pi}}_k\Big) = 2\boldsymbol{I}_k({}_j\!\times\dot{\boldsymbol{L}}_i) + \mathcal{B}_{k,\boldsymbol{V}_k}({}_j\!\times\boldsymbol{L}_i) + \mathcal{B}_{k,\dot{\boldsymbol{L}}_j}(\boldsymbol{L}_i), \tag{74}$$

$$\frac{\partial^2\dot{\mathbf{\Pi}}_k}{\partial\dot{q}_i\,\partial\dot{q}_j} = \mathcal{B}_{k,\boldsymbol{L}_j}(\boldsymbol{L}_i), \tag{75}$$

$$\mathcal{D}_i\left(\frac{\partial\dot{\mathbf{\Pi}}_k}{\partial\ddot{q}_j}\right) = 0, \tag{76}$$

$$\frac{\partial}{\partial\ddot{q}_i}\Big(\mathcal{D}_j\dot{\mathbf{\Pi}}_k\Big) = \boldsymbol{I}_k(\boldsymbol{L}_j \times \boldsymbol{L}_i), \tag{77}$$

$$\frac{\partial}{\partial\ddot{q}_i}\left(\frac{\partial\dot{\mathbf{\Pi}}_k}{\partial\dot{q}_j}\right) = 0, \tag{78}$$

$$\frac{\partial}{\partial\ddot{q}_i}\left(\frac{\partial\dot{\mathbf{\Pi}}_k}{\partial\ddot{q}_j}\right) = 0. \tag{79}$$

Eq. (72)

Proof. Start from (60):

$$\mathcal{D}_j\dot{\mathbf{\Pi}}_k = \boldsymbol{I}_k(\ddot{\boldsymbol{L}}_j) + \mathcal{B}_{k,\boldsymbol{V}_k}(\dot{\boldsymbol{L}}_j).$$

Apply $\mathcal{D}_i$:

$$\begin{aligned}\mathcal{D}_i\mathcal{D}_j\dot{\mathbf{\Pi}}_k &= \mathcal{D}_i(\boldsymbol{I}_k(\ddot{\boldsymbol{L}}_j)) + \mathcal{D}_i\big(\mathcal{B}_{k,\boldsymbol{V}_k}(\dot{\boldsymbol{L}}_j)\big)\\ &= \boldsymbol{I}_k(\mathcal{D}_i\ddot{\boldsymbol{L}}_j) + \mathcal{B}_{k,\boldsymbol{V}_k}(\mathcal{D}_i\dot{\boldsymbol{L}}_j) + \mathcal{B}_{k,\mathcal{D}_i\boldsymbol{V}_k}(\dot{\boldsymbol{L}}_j)\\ &= \boldsymbol{I}_k(2\dot{\boldsymbol{L}}_j \times \dot{\boldsymbol{L}}_i + \boldsymbol{L}_j \times \ddot{\boldsymbol{L}}_i) + \mathcal{B}_{k,\boldsymbol{V}_k}(\boldsymbol{L}_j \times \dot{\boldsymbol{L}}_i) + \mathcal{B}_{k,\dot{\boldsymbol{L}}_i}(\dot{\boldsymbol{L}}_j).\end{aligned} \tag{80}$$

Now use (55) with $\boldsymbol{X} = \dot{\boldsymbol{L}}_j$ and $Y = \dot{\boldsymbol{L}}_i$:

$$\mathcal{B}_{k,\dot{\boldsymbol{L}}_i}(\dot{\boldsymbol{L}}_j) + 2\boldsymbol{I}_k(\dot{\boldsymbol{L}}_j \times \dot{\boldsymbol{L}}_i) = \mathcal{B}_{k,\dot{\boldsymbol{L}}_j}(\dot{\boldsymbol{L}}_i).$$

Substituting this into (80) gives (72). □

Eq. (73)

Proof. Start from (61):

$$\frac{\partial\dot{\mathbf{\Pi}}_k}{\partial\dot{q}_j} = 2\boldsymbol{I}_k(\dot{\boldsymbol{L}}_j) + \mathcal{B}_{k,\boldsymbol{V}_k}(\boldsymbol{L}_j).$$

Apply $\mathcal{D}_i$:

$$\begin{aligned}\mathcal{D}_i\left(\frac{\partial\dot{\boldsymbol{\Pi}}_k}{\partial\dot{q}_j}\right) &= \mathcal{D}_i\big(\mathcal{B}_{k,\boldsymbol{V}_k}(\boldsymbol{L}_j)\big) + 2\mathcal{D}_i(\boldsymbol{I}_k(\dot{\boldsymbol{L}}_j)) \\ &= \mathcal{B}_{k,\boldsymbol{V}_k}(\mathcal{D}_i\boldsymbol{L}_j) + \mathcal{B}_{k,\mathcal{D}_i\boldsymbol{V}_k}(\boldsymbol{L}_j) + 2\boldsymbol{I}_k(\mathcal{D}_i\dot{\boldsymbol{L}}_j) \\ &= 0 + \mathcal{B}_{k,\dot{\boldsymbol{L}}_i}(\boldsymbol{L}_j) + 2\boldsymbol{I}_k(\boldsymbol{L}_j \times \dot{\boldsymbol{L}}_i). \end{aligned} \tag{81}$$

Now use (55) with $\boldsymbol{a} = \boldsymbol{L}_i$ and $\boldsymbol{b} = \dot{\boldsymbol{L}}_j$:

$$\mathcal{B}_{k,\dot{\boldsymbol{L}}_i}(\boldsymbol{L}_j) + 2\boldsymbol{I}_k(\boldsymbol{L}_j \times \dot{\boldsymbol{L}}_i) = \mathcal{B}_{k,\boldsymbol{L}_j}(\dot{\boldsymbol{L}}_i),$$

which proves (73). □

Eq. (74)

Proof. For (74), apply $\partial/\partial\dot{q}_i$ to (60):

$$\begin{aligned}\frac{\partial}{\partial\dot{q}_i}\Big(\mathcal{D}_j\dot{\boldsymbol{\Pi}}_k\Big) &= \boldsymbol{I}_k\left(\frac{\partial\ddot{\boldsymbol{L}}_j}{\partial\dot{q}_i}\right) + \frac{\partial}{\partial\dot{q}_i}\big(\mathcal{B}_{k,\boldsymbol{V}_k}(\dot{\boldsymbol{L}}_j)\big) \\ &= \boldsymbol{I}_k\Big(2(\dot{\boldsymbol{L}}_j \times \boldsymbol{L}_i + \boldsymbol{L}_j \times \dot{\boldsymbol{L}}_i)\Big) + \mathcal{B}_{k,\boldsymbol{V}_k}\left(\frac{\partial\dot{\boldsymbol{L}}_j}{\partial\dot{q}_i}\right) + \mathcal{B}_{k,\frac{\partial\boldsymbol{V}_k}{\partial\dot{q}_i}}(\dot{\boldsymbol{L}}_j) \\ &= 2\boldsymbol{I}_k(\dot{\boldsymbol{L}}_j \times \boldsymbol{L}_i + \boldsymbol{L}_j \times \dot{\boldsymbol{L}}_i) + \mathcal{B}_{k,\boldsymbol{V}_k}(\boldsymbol{L}_j \times \boldsymbol{L}_i) + \mathcal{B}_{k,\boldsymbol{L}_i}(\dot{\boldsymbol{L}}_j). \end{aligned} \tag{82}$$

Use (54) with $\boldsymbol{a} = \dot{\boldsymbol{L}}_j$ and $\boldsymbol{b} = \boldsymbol{L}_i$:

$$\mathcal{B}_{k,\dot{\boldsymbol{L}}_j}(\boldsymbol{L}_i) - \mathcal{B}_{k,\boldsymbol{L}_i}(\dot{\boldsymbol{L}}_j) = 2\boldsymbol{I}_k(\dot{\boldsymbol{L}}_j \times \boldsymbol{L}_i).$$

Hence

$$\mathcal{B}_{k,\boldsymbol{L}_i}(\dot{\boldsymbol{L}}_j) + 2\boldsymbol{I}_k(\dot{\boldsymbol{L}}_j \times \boldsymbol{L}_i) = \mathcal{B}_{k,\dot{\boldsymbol{L}}_j}(\boldsymbol{L}_i).$$

Substituting this into (82) proves (74). □

Eq. (75)

Proof. For (75), apply $\partial/\partial\dot{q}_i$ to (61):

$$\begin{aligned}\frac{\partial^2\dot{\boldsymbol{\Pi}}_k}{\partial\dot{q}_i\,\partial\dot{q}_j} &= \frac{\partial}{\partial\dot{q}_i}\big(\mathcal{B}_{k,\boldsymbol{V}_k}(\boldsymbol{L}_j)\big) + 2\boldsymbol{I}_k\left(\frac{\partial\dot{\boldsymbol{L}}_j}{\partial\dot{q}_i}\right) \\ &= \mathcal{B}_{k,\boldsymbol{V}_k}\left(\frac{\partial\boldsymbol{L}_j}{\partial\dot{q}_i}\right) + \mathcal{B}_{k,\frac{\partial\boldsymbol{V}_k}{\partial\dot{q}_i}}(\boldsymbol{L}_j) + 2\boldsymbol{I}_k(\boldsymbol{L}_j \times \boldsymbol{L}_i) \\ &= 0 + \mathcal{B}_{k,\boldsymbol{L}_i}(\boldsymbol{L}_j) + 2\boldsymbol{I}_k(\boldsymbol{L}_j \times \boldsymbol{L}_i). \end{aligned} \tag{83}$$

Using (55) with $\boldsymbol{a} = \boldsymbol{L}_j$ and $\boldsymbol{b} = \boldsymbol{L}_i$ yields

$$\mathcal{B}_{k,\boldsymbol{L}_i}(\boldsymbol{L}_j) + 2\boldsymbol{I}_k(\boldsymbol{L}_j \times \boldsymbol{L}_i) = \mathcal{B}_{k,\boldsymbol{L}_j}(\boldsymbol{L}_i),$$

which proves (75). □

Eq. (76)

Proof. For (76), start from (62):

$$\mathcal{D}_i\left(\frac{\partial \dot{\mathbf{\Pi}}_k}{\partial \ddot{q}_j}\right) = \mathcal{D}_i(\boldsymbol{I}_k(\boldsymbol{L}_j)) = \boldsymbol{I}_k(\mathcal{D}_i\boldsymbol{L}_j) = 0. \tag{84}$$

□

Eq. (77)

Proof. For (77), apply $\partial/\partial\ddot{q}_i$ to (60):

$$\begin{aligned}\frac{\partial}{\partial \ddot{q}_i}\left(\mathcal{D}_j\dot{\mathbf{\Pi}}_k\right) &= \boldsymbol{I}_k\left(\frac{\partial \ddot{\boldsymbol{L}}_j}{\partial \ddot{q}_i}\right) + \frac{\partial}{\partial \ddot{q}_i}\left(\mathcal{B}_{k,\boldsymbol{V}_k}(\dot{\boldsymbol{L}}_j)\right)\\ &= \boldsymbol{I}_k(\boldsymbol{L}_j \times \boldsymbol{L}_i) + \mathcal{B}_{k,\boldsymbol{V}_k}(0) + \mathcal{B}_{k,0}(\dot{\boldsymbol{L}}_j)\\ &= \boldsymbol{I}_k(\boldsymbol{L}_j \times \boldsymbol{L}_i).\end{aligned} \tag{85}$$

□

Eq. (78)

Proof. For (78), differentiate (61) with respect to $\ddot{q}_i$:

$$\frac{\partial}{\partial \ddot{q}_i}\left(\frac{\partial \dot{\mathbf{\Pi}}_k}{\partial \dot{q}_j}\right) = \frac{\partial}{\partial \ddot{q}_i}\left(\mathcal{B}_{k,\boldsymbol{V}_k}(\boldsymbol{L}_j)\right) + 2\boldsymbol{I}_k\left(\frac{\partial \dot{\boldsymbol{L}}_j}{\partial \ddot{q}_i}\right) = 0. \tag{86}$$

□

Eq. (79)

Proof. For (79), differentiate (62) with respect to $\ddot{q}_i$:

$$\frac{\partial}{\partial \ddot{q}_i}\left(\frac{\partial \dot{\mathbf{\Pi}}_k}{\partial \ddot{q}_j}\right) = \frac{\partial}{\partial \ddot{q}_i}\boldsymbol{I}_k(\boldsymbol{L}_j) = 0. \tag{87}$$

This completes the proof. □

3.2 Second-order derivatives of τ_k

The SO Derivative of τ With Respect to q

$$\begin{aligned}\frac{\partial^2 \tau_k}{\partial q_i \partial q_j} &= \frac{\partial}{\partial q_i}\left(\boldsymbol{L}_k * \sum_{m \succeq k}(\mathcal{D}_j\dot{\mathbf{\Pi}}_m)\right)\\ &= \boldsymbol{L}_k * \sum_{m \succeq k}\left(\boldsymbol{L}_i \times (\mathcal{D}_j\dot{\mathbf{\Pi}}_m) + \frac{\partial}{\partial q_i}(\mathcal{D}_j\dot{\mathbf{\Pi}}_m)\right)\\ &= \boldsymbol{L}_k * \sum_{m \succeq k}(\mathcal{D}_i\mathcal{D}_j\dot{\mathbf{\Pi}}_m)\\ &= \boldsymbol{L}_k * \sum_{m \succeq k}\left(\boldsymbol{I}_m(\boldsymbol{L}_j \times \ddot{\boldsymbol{L}}_i) + \mathcal{B}_{m,\boldsymbol{V}_m}(\boldsymbol{L}_j \times \dot{\boldsymbol{L}}_i) + \mathcal{B}_{m,\dot{\boldsymbol{L}}_j}(\dot{\boldsymbol{L}}_i)\right)\\ &= \boldsymbol{L}_k * \left(\boldsymbol{I}_k^C(\boldsymbol{L}_j \times \ddot{\boldsymbol{L}}_i) + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\boldsymbol{L}_j \times \dot{\boldsymbol{L}}_i) + \mathcal{B}^C_{k,\dot{\boldsymbol{L}}_j}(\dot{\boldsymbol{L}}_i)\right).\end{aligned} \tag{88}$$

$$\frac{\partial^2 \tau_k}{\partial q_j \partial q_i} = \frac{\partial^2 \tau_k}{\partial q_i \partial q_j} \tag{89}$$

$$\begin{aligned}
\frac{\partial^2 \tau_j}{\partial q_i \partial q_k} &= \frac{\partial}{\partial q_i}\left(\boldsymbol{L}_j * \sum_{m \succeq k} (\mathcal{D}_k \dot{\boldsymbol{\Pi}}_m + \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k)\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} \left(\boldsymbol{L}_i \times (\mathcal{D}_k \dot{\boldsymbol{\Pi}}_m + \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k) + \frac{\partial}{\partial q_i}(\mathcal{D}_k \dot{\boldsymbol{\Pi}}_m + \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k)\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} \left(\mathcal{D}_i \mathcal{D}_k \dot{\boldsymbol{\Pi}}_m + \mathcal{D}_i(\dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k)\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} \left(\boldsymbol{I}_m(\boldsymbol{L}_k \times \ddot{\boldsymbol{L}}_i) + \mathcal{B}_{m,\boldsymbol{V}_m}(\boldsymbol{L}_k \times \dot{\boldsymbol{L}}_i) + \mathcal{B}_{m,\dot{\boldsymbol{L}}_k}(\dot{\boldsymbol{L}}_i) + \mathcal{D}_i \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} \left(\boldsymbol{I}_m(\boldsymbol{L}_k \times \ddot{\boldsymbol{L}}_i) + \mathcal{B}_{m,\boldsymbol{V}_m}(\boldsymbol{L}_k \times \dot{\boldsymbol{L}}_i) + \mathcal{B}_{m,\dot{\boldsymbol{L}}_k}(\dot{\boldsymbol{L}}_i) + \boldsymbol{I}_m(\ddot{\boldsymbol{L}}_i) \times \boldsymbol{L}_k + \mathcal{B}_{m,\boldsymbol{V}_m}(\dot{\boldsymbol{L}}_i) \times \boldsymbol{L}_k\right) \\
&= \boldsymbol{L}_j * \left(\boldsymbol{I}_k^C(\boldsymbol{L}_k \times \ddot{\boldsymbol{L}}_i) + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\boldsymbol{L}_k \times \dot{\boldsymbol{L}}_i) + \mathcal{B}^C_{k,\dot{\boldsymbol{L}}_k}(\dot{\boldsymbol{L}}_i) + \boldsymbol{I}_k^C(\ddot{\boldsymbol{L}}_i) \times \boldsymbol{L}_k + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\dot{\boldsymbol{L}}_i) \times \boldsymbol{L}_k\right)
\end{aligned} \tag{90}$$

$$\frac{\partial^2 \tau_j}{\partial q_k \partial q_i} = \frac{\partial^2 \tau_j}{\partial q_i \partial q_k} \tag{91}$$

$$\begin{aligned}
\frac{\partial^2 \tau_i}{\partial q_j \partial q_k} &= \frac{\partial}{\partial q_j}\left(\boldsymbol{L}_i * \sum_{m \succeq k} (\mathcal{D}_k \dot{\boldsymbol{\Pi}}_m + \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k)\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} \left(\frac{\partial}{\partial q_j}(\mathcal{D}_k \dot{\boldsymbol{\Pi}}_m + \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k)\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} \left(\mathcal{D}_j \mathcal{D}_k \dot{\boldsymbol{\Pi}}_m + \mathcal{D}_k \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_j + \mathcal{D}_j(\dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k) + (\dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k) \times \boldsymbol{L}_j\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} \left(\mathcal{D}_j \mathcal{D}_k \dot{\boldsymbol{\Pi}}_m + \mathcal{D}_k \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_j + \mathcal{D}_j \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k + (\dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k) \times \boldsymbol{L}_j\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} \Big(\boldsymbol{I}_m(\boldsymbol{L}_k \times \ddot{\boldsymbol{L}}_j) + \mathcal{B}_{m,\boldsymbol{V}_m}(\boldsymbol{L}_k \times \dot{\boldsymbol{L}}_j) + \mathcal{B}_{m,\dot{\boldsymbol{L}}_k}(\dot{\boldsymbol{L}}_j) + \boldsymbol{I}_m(\ddot{\boldsymbol{L}}_j) \times \boldsymbol{L}_k \\
&\quad + \mathcal{B}_{m,\boldsymbol{V}_m}(\dot{\boldsymbol{L}}_j) \times \boldsymbol{L}_k + \boldsymbol{I}_m(\ddot{\boldsymbol{L}}_k) \times \boldsymbol{L}_j + \mathcal{B}_{m,\boldsymbol{V}_m}(\dot{\boldsymbol{L}}_k) \times \boldsymbol{L}_j + (\dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k) \times \boldsymbol{L}_j\Big) \\
&= \boldsymbol{L}_i * \Big(\boldsymbol{I}_k^C(\boldsymbol{L}_k \times \ddot{\boldsymbol{L}}_j) + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\boldsymbol{L}_k \times \dot{\boldsymbol{L}}_j) + \mathcal{B}^C_{k,\dot{\boldsymbol{L}}_k}(\dot{\boldsymbol{L}}_j) + \boldsymbol{I}_k^C(\ddot{\boldsymbol{L}}_j) \times \boldsymbol{L}_k \\
&\quad + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\dot{\boldsymbol{L}}_j) \times \boldsymbol{L}_k + \boldsymbol{I}_k^C(\ddot{\boldsymbol{L}}_k) \times \boldsymbol{L}_j + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\dot{\boldsymbol{L}}_k) \times \boldsymbol{L}_j + (\dot{\boldsymbol{\Pi}}_k^C \times \boldsymbol{L}_k) \times \boldsymbol{L}_j\Big)
\end{aligned} \tag{92}$$

$$\frac{\partial^2 \tau_i}{\partial q_k \partial q_j} = \frac{\partial^2 \tau_i}{\partial q_j \partial q_k} \tag{93}$$

Cross SO Derivative of τ With Respect to q and $\dot{q}$

$$\begin{aligned}
\frac{\partial^2 \tau_k}{\partial \dot{q}_i \partial q_j} &= \frac{\partial}{\partial \dot{q}_i}\left(\boldsymbol{L}_k * \sum_{m \succeq k} (\mathcal{D}_j \dot{\boldsymbol{\Pi}}_m)\right) \\
&= \boldsymbol{L}_k * \sum_{m \succeq k} \left(\frac{\partial}{\partial \dot{q}_i}(\mathcal{D}_j \dot{\boldsymbol{\Pi}}_m)\right) \\
&= \boldsymbol{L}_k * \sum_{m \succeq k} \Big(2\boldsymbol{I}_m(\boldsymbol{L}_j \times \dot{\boldsymbol{L}}_i) + \mathcal{B}_{m,\boldsymbol{V}_m}(\boldsymbol{L}_j \times \boldsymbol{L}_i) + \mathcal{B}_{m,\dot{\boldsymbol{L}}_j}(\boldsymbol{L}_i)\Big) \\
&= \boldsymbol{L}_k * \Big(2\boldsymbol{I}_k^C(\boldsymbol{L}_j \times \dot{\boldsymbol{L}}_i) + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\boldsymbol{L}_j \times \boldsymbol{L}_i) + \mathcal{B}^C_{k,\dot{\boldsymbol{L}}_j}(\boldsymbol{L}_i)\Big)
\end{aligned} \tag{94}$$

$$\begin{aligned}
\frac{\partial^2 \tau_k}{\partial \dot{q}_j \partial q_i} &= \frac{\partial}{\partial \dot{q}_j}\left(\boldsymbol{L}_k * \sum_{m \succeq k} (\mathcal{D}_i \dot{\boldsymbol{\Pi}}_m)\right) \\
&= \boldsymbol{L}_k * \sum_{m \succeq k} \left(\frac{\partial}{\partial \dot{q}_j}(\mathcal{D}_i \dot{\boldsymbol{\Pi}}_m)\right) \\
&= \boldsymbol{L}_k * \sum_{m \succeq k} \Big(\mathcal{B}_{m,\boldsymbol{L}_j}(\dot{\boldsymbol{L}}_i)\Big) \\
&= \boldsymbol{L}_k * \mathcal{B}^C_{k,\boldsymbol{L}_j}(\dot{\boldsymbol{L}}_i)
\end{aligned} \tag{95}$$

$$\begin{aligned}
\frac{\partial^2 \tau_j}{\partial \dot{q}_i \partial q_k} &= \frac{\partial}{\partial \dot{q}_i}\left(\boldsymbol{L}_j * \sum_{m \succeq k} (\mathcal{D}_k \dot{\boldsymbol{\Pi}}_m + \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k)\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} \left(\frac{\partial}{\partial \dot{q}_i}(\mathcal{D}_k \dot{\boldsymbol{\Pi}}_m) + \frac{\partial}{\partial \dot{q}_i}\dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} \Big(2\boldsymbol{I}_m(\boldsymbol{L}_k \times \dot{\boldsymbol{L}}_i) + \mathcal{B}_{m,\boldsymbol{V}_m}(\boldsymbol{L}_k \times \boldsymbol{L}_i) + \mathcal{B}_{m,\dot{\boldsymbol{L}}_k}(\boldsymbol{L}_i) + 2\boldsymbol{I}_m(\dot{\boldsymbol{L}}_i) \times \boldsymbol{L}_k + \mathcal{B}_{m,\boldsymbol{V}_m}(\boldsymbol{L}_i) \times \boldsymbol{L}_k\Big) \\
&= \boldsymbol{L}_j * \Big(2\boldsymbol{I}_k^C(\boldsymbol{L}_k \times \dot{\boldsymbol{L}}_i) + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\boldsymbol{L}_k \times \boldsymbol{L}_i) + \mathcal{B}^C_{k,\dot{\boldsymbol{L}}_k}(\boldsymbol{L}_i) + 2\boldsymbol{I}_k^C(\dot{\boldsymbol{L}}_i) \times \boldsymbol{L}_k + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\boldsymbol{L}_i) \times \boldsymbol{L}_k\Big)
\end{aligned} \tag{96}$$

$$\begin{aligned}
\frac{\partial^2 \tau_j}{\partial \dot{q}_k \partial q_i} &= \frac{\partial}{\partial \dot{q}_k}\left(\boldsymbol{L}_j * \sum_{m \succeq k} (\mathcal{D}_i \dot{\boldsymbol{\Pi}}_m)\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} \left(\frac{\partial}{\partial \dot{q}_k}(\mathcal{D}_i \dot{\boldsymbol{\Pi}}_m)\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} \Big(\mathcal{B}_{m,\boldsymbol{L}_k}(\dot{\boldsymbol{L}}_i)\Big) \\
&= \boldsymbol{L}_j * \mathcal{B}^C_{k,\boldsymbol{L}_k}(\dot{\boldsymbol{L}}_i)
\end{aligned} \tag{97}$$

$$
\begin{aligned}
\frac{\partial^2 \tau_i}{\partial \dot{q}_j \partial q_k} &= \frac{\partial}{\partial \dot{q}_j}\left(\boldsymbol{L}_i * \sum_{m \succeq k} (\mathcal{D}_k \dot{\boldsymbol{\Pi}}_m + \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k)\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k}\left(\frac{\partial}{\partial \dot{q}_j}(\mathcal{D}_k \dot{\boldsymbol{\Pi}}_m) + \frac{\partial}{\partial \dot{q}_j}\dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_k\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k}\Big(2\boldsymbol{I}_m(\boldsymbol{L}_k \times \dot{\boldsymbol{L}}_j) + \mathcal{B}_{m,\boldsymbol{V}_m}(\boldsymbol{L}_k \times \boldsymbol{L}_j) + \mathcal{B}_{m,\dot{\boldsymbol{L}}_k}(\boldsymbol{L}_j) + 2\boldsymbol{I}_m(\dot{\boldsymbol{L}}_j) \times \boldsymbol{L}_k + \mathcal{B}_{m,\boldsymbol{V}_m}(\boldsymbol{L}_j) \times \boldsymbol{L}_k\Big) \\
&= \boldsymbol{L}_i * \Big(2\boldsymbol{I}_k^C(\boldsymbol{L}_k \times \dot{\boldsymbol{L}}_j) + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\boldsymbol{L}_k \times \boldsymbol{L}_j) + \mathcal{B}^C_{k,\dot{\boldsymbol{L}}_k}(\boldsymbol{L}_j) + 2\boldsymbol{I}_k^C(\dot{\boldsymbol{L}}_j) \times \boldsymbol{L}_k + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\boldsymbol{L}_j) \times \boldsymbol{L}_k\Big)
\end{aligned} \tag{98}
$$

$$
\begin{aligned}
\frac{\partial^2 \tau_i}{\partial \dot{q}_k \partial q_j} &= \frac{\partial}{\partial \dot{q}_k}\left(\boldsymbol{L}_i * \sum_{m \succeq k} (\mathcal{D}_j \dot{\boldsymbol{\Pi}}_m + \dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_j)\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k}\left(\frac{\partial}{\partial \dot{q}_k}(\mathcal{D}_j \dot{\boldsymbol{\Pi}}_m) + \frac{\partial}{\partial \dot{q}_k}\dot{\boldsymbol{\Pi}}_m \times \boldsymbol{L}_j\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k}\Big(\mathcal{B}_{m,\boldsymbol{L}_k}(\dot{\boldsymbol{L}}_j) + 2\boldsymbol{I}_m(\dot{\boldsymbol{L}}_k) \times \boldsymbol{L}_j + \mathcal{B}_{m,\boldsymbol{V}_m}(\boldsymbol{L}_k) \times \boldsymbol{L}_j\Big) \\
&= \boldsymbol{L}_i * \Big(\mathcal{B}^C_{k,\boldsymbol{L}_k}(\dot{\boldsymbol{L}}_j) + 2\boldsymbol{I}_k^C(\dot{\boldsymbol{L}}_k) \times \boldsymbol{L}_j + \mathcal{B}^C_{k,\boldsymbol{V}_k}(\boldsymbol{L}_k) \times \boldsymbol{L}_j\Big)
\end{aligned} \tag{99}
$$

The SO Derivative of τ With Respect to $\dot{q}$

$$
\begin{aligned}
\frac{\partial^2 \tau_k}{\partial \dot{q}_i \partial \dot{q}_j} &= \frac{\partial}{\partial \dot{q}_i}\left(\boldsymbol{L}_k * \sum_{m \succeq k} (\frac{\partial}{\partial \dot{q}_j}\dot{\boldsymbol{\Pi}}_m)\right) \\
&= \boldsymbol{L}_k * \sum_{m \succeq k}\left(\frac{\partial^2 \dot{\boldsymbol{\Pi}}_m}{\partial \dot{q}_i \partial \dot{q}_j}\right) \\
&= \boldsymbol{L}_k * \sum_{m \succeq k}\big(\mathcal{B}_{m,\boldsymbol{L}_j}(\boldsymbol{L}_i)\big) \\
&= \boldsymbol{L}_k * \mathcal{B}^C_{k,\boldsymbol{L}_j}(\boldsymbol{L}_i)
\end{aligned} \tag{100}
$$

$$
\frac{\partial^2 \tau_k}{\partial \dot{q}_j \partial \dot{q}_i} = \frac{\partial^2 \tau_k}{\partial \dot{q}_i \partial \dot{q}_j} \tag{101}
$$

$$
\begin{aligned}
\frac{\partial^2 \tau_j}{\partial \dot{q}_i \partial \dot{q}_k} &= \frac{\partial}{\partial \dot{q}_i}\left(\boldsymbol{L}_j * \sum_{m \succeq k} (\frac{\partial}{\partial \dot{q}_k}\dot{\boldsymbol{\Pi}}_m)\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k}\left(\frac{\partial^2 \dot{\boldsymbol{\Pi}}_m}{\partial \dot{q}_i \partial \dot{q}_k}\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k}(\mathcal{B}_{m,\boldsymbol{L}_k}(\boldsymbol{L}_i)) \\
&= \boldsymbol{L}_j * \mathcal{B}^C_{k,\boldsymbol{L}_k}(\boldsymbol{L}_i)
\end{aligned} \tag{102}
$$

$$\frac{\partial^2 \tau_j}{\partial \dot{q}_k \partial \dot{q}_i} = \frac{\partial^2 \tau_j}{\partial \dot{q}_i \partial \dot{q}_k} \tag{103}$$

$$\begin{aligned}
\frac{\partial^2 \tau_i}{\partial \dot{q}_j \partial \dot{q}_k} &= \frac{\partial}{\partial \dot{q}_j} \left(\boldsymbol{L}_i * \sum_{m \succeq k} (\frac{\partial}{\partial \dot{q}_k} \dot{\boldsymbol{\Pi}}_m) \right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} \left(\frac{\partial^2 \dot{\boldsymbol{\Pi}}_m}{\partial \dot{q}_j \partial \dot{q}_k} \right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} (\mathcal{B}_{m, \boldsymbol{L}_k}(\boldsymbol{L}_j)) \\
&= \boldsymbol{L}_i * \mathcal{B}^C_{k, \boldsymbol{L}_k}(\boldsymbol{L}_j)
\end{aligned} \tag{104}$$

$$\frac{\partial^2 \tau_i}{\partial \dot{q}_k \partial \dot{q}_j} = \frac{\partial^2 \tau_i}{\partial \dot{q}_j \partial \dot{q}_k} \tag{105}$$

Cross SO Derivative of τ With Respect to q and $\ddot{q}$

$$\begin{aligned}
\frac{\partial^2 \tau_k}{\partial q_i \partial \ddot{q}_j} &= \frac{\partial}{\partial q_i} \left(\boldsymbol{L}_k * \sum_{m \succeq k} (\frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_j}) \right) \\
&= \boldsymbol{L}_k * \sum_{m \succeq k} \left(\boldsymbol{L}_i \times \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_j} + \frac{\partial}{\partial q_i} \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_j} \right) \\
&= \boldsymbol{L}_k * \sum_{m \succeq k} \left(\mathcal{D}_i \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_j} \right) \\
&= \boldsymbol{L}_k * \sum_{m \succeq k} (0) = 0
\end{aligned} \tag{106}$$

$$\begin{aligned}
\frac{\partial^2 \tau_k}{\partial q_j \partial \ddot{q}_i} &= \frac{\partial}{\partial q_j} \left(\boldsymbol{L}_k * \sum_{m \succeq k} (\frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_i}) \right) \\
&= \boldsymbol{L}_k * \sum_{m \succeq k} \left(\mathcal{D}_j \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_i} \right) \\
&= \boldsymbol{L}_k * \sum_{m \succeq k} (\boldsymbol{I}_m (\boldsymbol{L}_j \times \boldsymbol{L}_i)) \\
&= \boldsymbol{L}_k * \boldsymbol{I}^C_k (\boldsymbol{L}_j \times \boldsymbol{L}_i)
\end{aligned} \tag{107}$$

$$
\begin{aligned}
\frac{\partial^2 \tau_j}{\partial q_i \partial \ddot{q}_k} &= \frac{\partial}{\partial q_i}\left(\boldsymbol{L}_j * \sum_{m \succeq k} (\frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_k})\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} \left(\mathcal{D}_i \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_k}\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} (0) = 0
\end{aligned} \tag{108}
$$

$$
\begin{aligned}
\frac{\partial^2 \tau_j}{\partial q_k \partial \ddot{q}_i} &= \frac{\partial}{\partial q_k}\left(\boldsymbol{L}_j * \sum_{m \succeq k} (\frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_i})\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} \left(\frac{\partial}{\partial q_k} \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_i}\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} \left(\mathcal{D}_k \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_i} + \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_i} \times \boldsymbol{L}_k\right) \\
&= \boldsymbol{L}_j * \sum_{m \succeq k} (\boldsymbol{I}_m(\boldsymbol{L}_k \times \boldsymbol{L}_i) + \boldsymbol{I}_m(\boldsymbol{L}_i) \times \boldsymbol{L}_k) \\
&= \boldsymbol{L}_j * \left(\boldsymbol{I}_k^C(\boldsymbol{L}_k \times \boldsymbol{L}_i) + \boldsymbol{I}_k^C(\boldsymbol{L}_i) \times \boldsymbol{L}_k\right)
\end{aligned} \tag{109}
$$

$$
\begin{aligned}
\frac{\partial^2 \tau_i}{\partial q_j \partial \ddot{q}_k} &= \frac{\partial}{\partial q_j}\left(\boldsymbol{L}_i * \sum_{m \succeq k} (\frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_k})\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} \left(\mathcal{D}_j \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_k} + \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_k} \times \boldsymbol{L}_j\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} (0 + \boldsymbol{I}_m(\boldsymbol{L}_k) \times \boldsymbol{L}_j) \\
&= \boldsymbol{L}_i * \boldsymbol{I}_k^C(\boldsymbol{L}_k) \times \boldsymbol{L}_j
\end{aligned} \tag{110}
$$

$$
\begin{aligned}
\frac{\partial^2 \tau_i}{\partial q_k \partial \ddot{q}_j} &= \frac{\partial}{\partial q_k}\left(\boldsymbol{L}_i * \sum_{m \succeq k} (\frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_j})\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} \left(\mathcal{D}_k \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_j} + \frac{\partial \dot{\boldsymbol{\Pi}}_m}{\partial \ddot{q}_j} \times \boldsymbol{L}_k\right) \\
&= \boldsymbol{L}_i * \sum_{m \succeq k} (\boldsymbol{I}_m(\boldsymbol{L}_k \times \boldsymbol{L}_j) + \boldsymbol{I}_m(\boldsymbol{L}_j) \times \boldsymbol{L}_k) \\
&= \boldsymbol{L}_i * \left(\boldsymbol{I}_k^C(\boldsymbol{L}_k \times \boldsymbol{L}_j) + \boldsymbol{I}_k^C(\boldsymbol{L}_j) \times \boldsymbol{L}_k\right)
\end{aligned} \tag{111}
$$

4 Derivative Correctness

We verify the correctness of the proposed analytical derivatives by comparing them against central finite-difference references (with step sizes of 5×10^{-7} for the first-order and 1×10^{-5} for the second-order derivatives) computed from the original inverse-dynamics function. We report the median and worst-case maximum absolute errors, over 10 randomly sampled states.

Table 1: Accuracy validation of first-order analytical inverse-dynamics derivatives on Franka Emika Panda.

Derivative item	Median rel. err.	Worst rel. err.	Median max abs. err.	Worst max abs. err.
$\partial\tau/\partial q$	5.55×10^{-11}	2.93×10^{-10}	1.177×10^{-9}	4.384×10^{-9}
$\partial\tau/\partial \dot{q}$	4.013×10^{-10}	2.207×10^{-9}	9.59×10^{-10}	3.093×10^{-9}
$\partial\tau/\partial \ddot{q}$	5.362×10^{-10}	1.311×10^{-9}	8.94×10^{-10}	3.889×10^{-9}

Table 2: Accuracy validation of second-order analytical inverse-dynamics derivatives on Franka Emika Panda.

Derivative item	Median rel. err.	Worst rel. err.	Median max abs. err.	Worst max abs. err.
$\partial^2\tau/\partial q^2$	4.55×10^{-8}	7.87×10^{-8}	1.353×10^{-6}	3.691×10^{-6}
$\partial^2\tau/\partial q\,\partial \dot{q}$	1.315×10^{-7}	5.233×10^{-7}	3.68×10^{-7}	8.24×10^{-7}
$\partial^2\tau/\partial \dot{q}^2$	4.435×10^{-7}	7.890×10^{-7}	9.92×10^{-7}	2.127×10^{-6}
$\partial^2\tau/\partial q\,\partial \ddot{q}$	3.081×10^{-7}	5.503×10^{-7}	3.30×10^{-7}	6.16×10^{-7}

Figure 1 plots the error between the analytical derivatives and the numerical results obtained via finite differences as the model's degrees of freedom increase. The results demonstrate that the analytical derivatives are in excellent agreement with the numerical reference values.

5 Additional Benchmark Results

To evaluate the hardware robustness of the proposed implementation, we conducted cross-CPU benchmarks on three representative computing platforms covering desktop, laptop, and embedded execution scenarios. The tested platforms include:

1. a desktop Intel Core i7-13700K system with 32 GB memory,
2. a laptop AMD Ryzen 7 6800H system with 32 GB memory,
3. a Raspberry Pi 4B equipped with an ARM Cortex-A72 processor and 8 GB memory.

These three platforms were selected to represent distinct deployment settings commonly encountered in robotics research and practice. The Intel desktop platform represents a high-performance local workstation, the AMD laptop platform represents a mobile x86 computing environment, and the Raspberry Pi platform represents a resource-constrained embedded ARM device. Rather than comparing processor vendors in isolation, the purpose of this benchmark is to assess whether the relative computational trends of the proposed method remain consistent across these different execution regimes.

On each platform, we benchmarked five computational tasks: dynamics evaluation, first-order dynamics derivatives, second-order dynamics derivatives, DDP solve time, and SQP solve

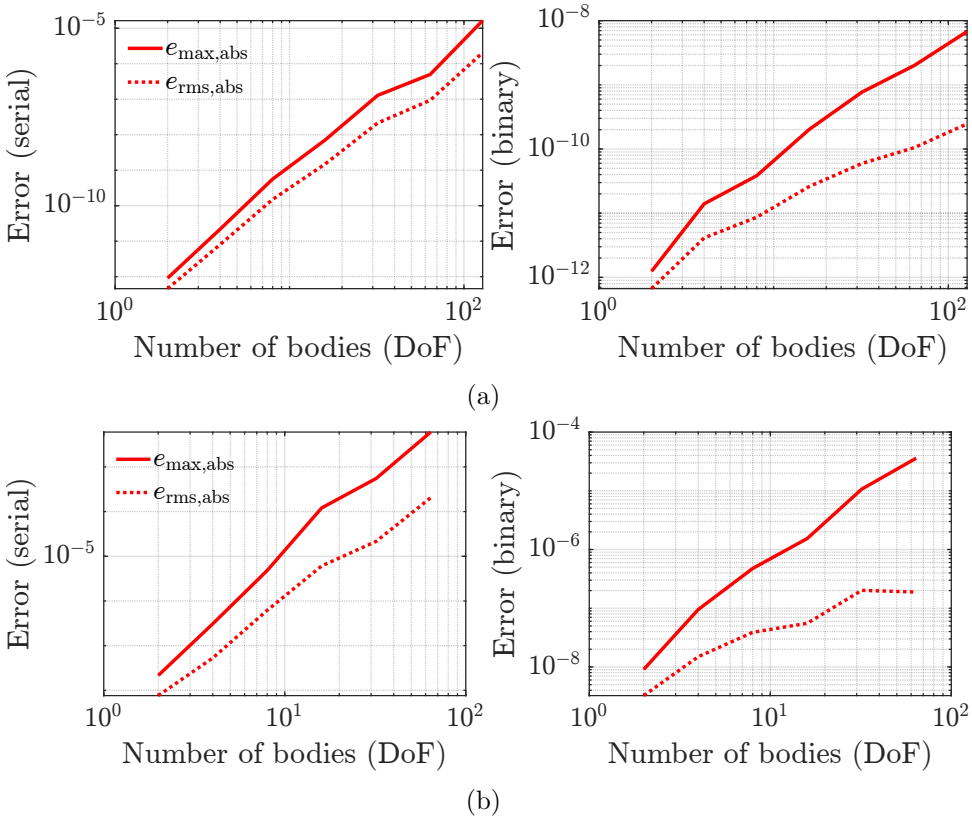

Figure 1: Derivative verification as a function of system dimension. (a) First-order derivative errors. (b) Second-order derivative errors. Results are reported for both serial chains and binary trees using central finite-difference references.

time. For each task, runtime was measured as a function of the robot degrees of freedom (DoFs), using the same DoF sweep and the same benchmark scripts across all platforms. The proposed implementation was compared against Pinocchio and CasADi under matched task definitions and solver settings.

Table 3: Hardware platforms used in the cross-CPU benchmarks.

Platform	CPU Model	Memory	Max Frequency	Architecture	Notes
Intel Desktop	Intel Core i7-13700K	32GB	5.3 GHz	x86_64	High-performance local machine
AMD Laptop	AMD Ryzen 7 6800H	32GB	4.6 GHz	x86_64	Mobile x86 laptop
Raspberry Pi 4B	Broadcom BCM2711, Cortex-A72	8GB	1.8 GHz	aarch64	Resource-constrained embedded platform

Unless otherwise stated, all reported runtimes correspond to CPU-only execution and were averaged over 256 repeated runs. The measurements were designed to reflect repeated evaluation during solver execution rather than one-time setup overhead such as model loading or graph construction. Detailed hardware specifications are summarized in Table 3.

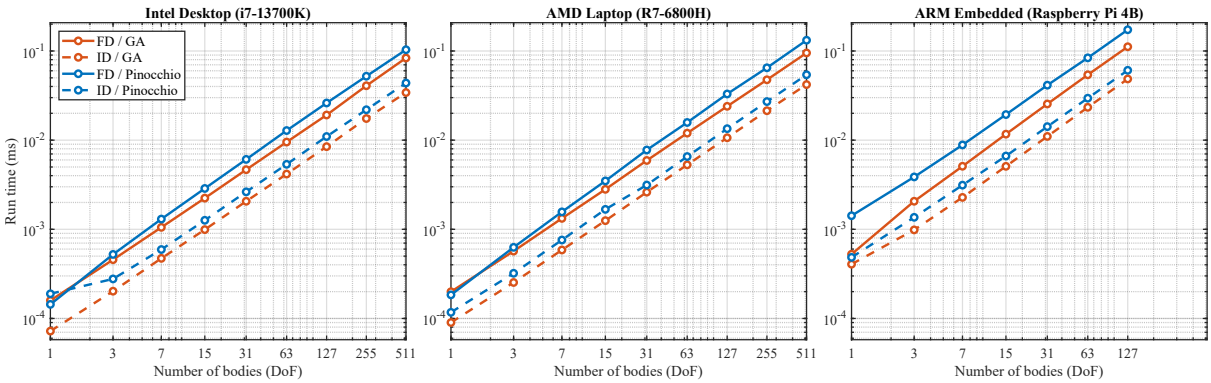

Figure 2: Cross-CPU scalability of dynamics versus robot DoFs on desktop, laptop, and embedded platforms. The proposed method is compared against Pinocchio under matched benchmark settings.

The cross-CPU results show that the absolute runtime varies substantially across the three tested platforms, as expected from their different hardware classes. However, the relative trends remain consistent: the proposed implementation remains competitive in dynamics evaluation and shows increasingly clear advantages as the derivative order and system dimension increase. This consistency across desktop, laptop, and embedded settings suggests that the computational benefits of the proposed formulation are not tied to a single hardware environment.

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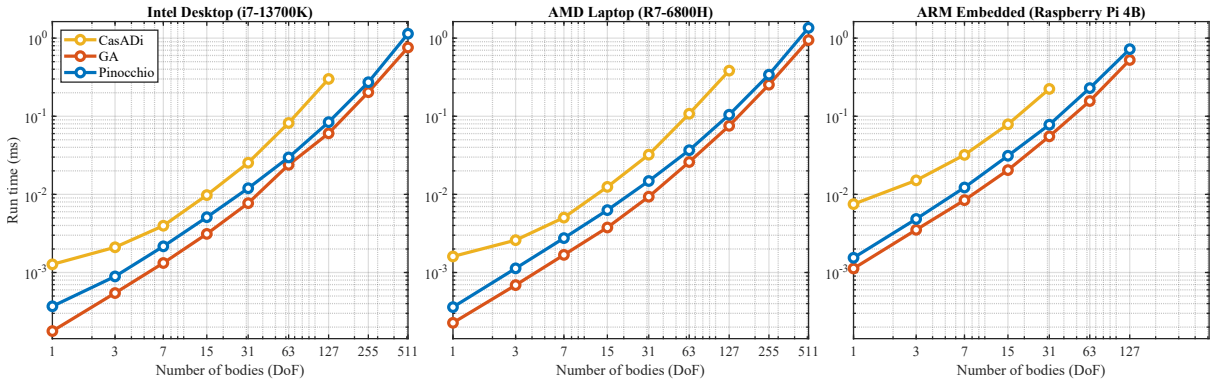

Figure 3: Cross-CPU scalability of first-order derivatives of ID versus robot DoFs on desktop, laptop, and embedded platforms. The proposed method is compared against Pinocchio and CasADi under matched benchmark settings.

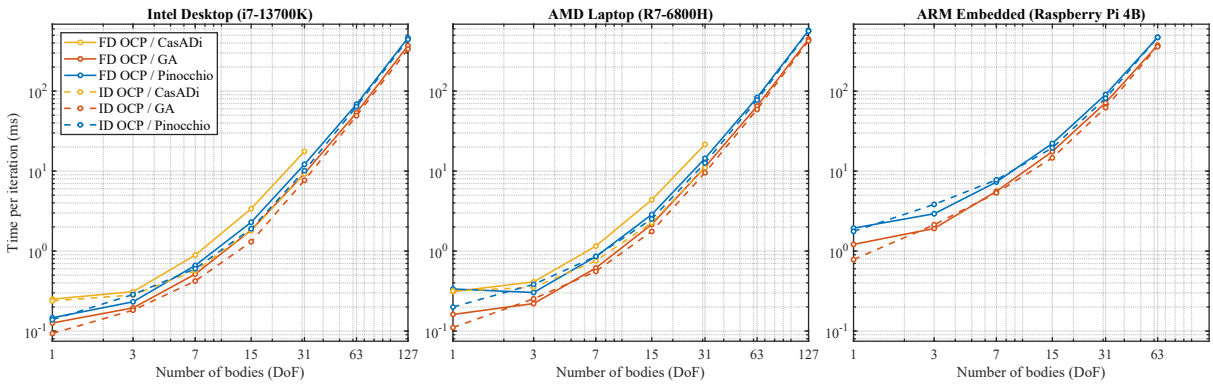

Figure 4: Cross-CPU scalability of Crocoddyl FDDP solver versus robot DoFs on desktop, laptop, and embedded platforms. The proposed method is compared against Pinocchio and CasADi under matched benchmark settings.

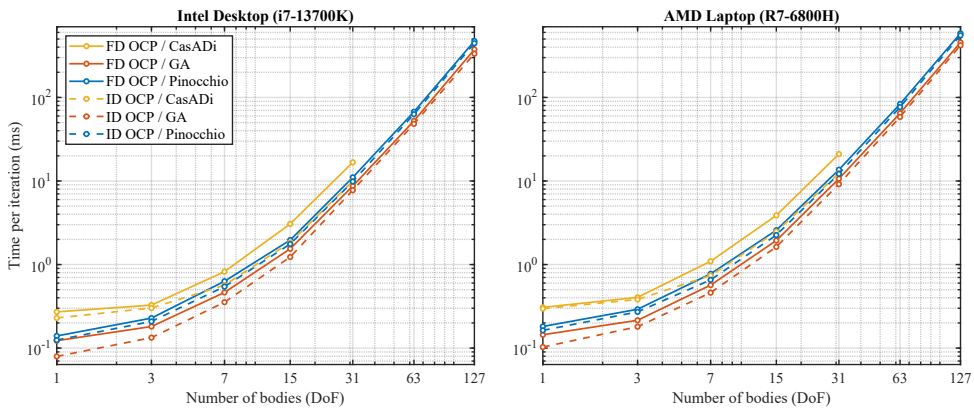

Figure 5: Cross-CPU scalability of StagewiseSQP solver versus robot DoFs on desktop and laptop. The proposed method is compared against Pinocchio and CasADi under matched benchmark settings.

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